

cars; a 5G-enabled fleet management platform for real-time tracking and predictive maintenance; and a General Perception Intelligence (GPI) framework that fuses vision, lidar, radar, and polarised light into a machine learning pipeline for detection and decision-making. RoshAi positions its offering as a vehicle- and sensor-agnostic autonomy operating system, supported by leadership with extensive experience in robotics and autonomous mobility on Indian roads.

Established automakers are advancing in parallel. Tata Motors, Mahindra, and Ashok Leyland continue to trial semi-autonomous and autonomous capabilities within controlled operational design domains such as mines, ports, and industrial campuses. Partnerships are also emerging. For example, Ashok Leyland's collaboration with Minus Zero on self-driving trucks for industrial settings signals a pragmatic, application-first route to autonomy in India.

India's IT majors, Infosys, TCS, and Wipro, serve as bridges between global original equipment manufacturers and Tier-1 suppliers and domestic pilot programmes. Their strengths in software platforms, validation and simulation, cybersecurity, and lifecycle support help adapt mature technologies to India's traffic realities and cost structures.

On the policy front, NITI Aayog's National Strategy for AI identifies mobility as a priority, while the Smart Cities Mission and broader electric vehicle incentives create favourable conditions for pilot projects. What matters next is the critical foundational infrastructure: clear testing standards and certification pathways, sandboxed trials, high-quality datasets, and high-definition map layers. Data-sharing frameworks with strong privacy safeguards are also required. Finally, incremental investment in connected infrastructure, reliable lane markings, calibrated signals, and V2X-ready corridors will be essential to enable autonomy to scale safely.

Safety, society, and regulation

India accounts for nearly 11% of global road fatalities while owning only around 1% of the world's vehicles. This disparity underscores the urgent need for safer mobility solutions. Autonomous and semi-autonomous systems have the potential to reduce this toll through features such as lane-keeping assistance, automatic emergency braking, drowsiness detection, adaptive cruise control, and collision avoidance. These technologies could not only save lives but also reduce the economic burden of accidents, which currently costs India an estimated 3-5% of its GDP each year.

At the same time, public perception remains a formidable barrier. For many drivers, ceding control to an AI-driven system raises concerns about trust and safety. Fears around job displacement, particularly among truck, taxi, and delivery drivers, further complicate acceptance.

However, this disruption is double-edged. As the ecosystem matures, new roles are expected to emerge in areas such as AI model training, sensor calibration, simulation environments, fleet monitoring, cybersecurity, and real-time remote support for autonomous fleets. For India's large pool of engineers and technicians, this shift presents an opportunity to reskill and participate in the future of mobility.

On the regulatory front, India still lacks a comprehensive framework for autonomous vehicles, although early steps are underway. NITI Aayog has emphasised mobility in its National Strategy on AI, and several state-level authorities have begun exploring controlled pilot deployments. Transparent trials in industrial campuses, ports, mines, and smart city corridors can act as stepping stones before broader public deployment. At the national level, policies must address safety certification, liability in the event of accidents, data privacy, and ethical considerations.

Ultimately, regulation must strike a careful balance between encouraging innovation and safeguarding lives and livelihoods. If India establishes robust standards and early pilot frameworks, it can accelerate domestic adoption and position itself as a global hub for affordable, safety-first autonomous solutions.

The road ahead: Semi-autonomy first

For India, the near-term future lies in semi-autonomous systems, driver-assistance technologies that enhance human capability rather than replace drivers entirely. These systems can act as a transitional bridge, reducing accidents and improving efficiency while preparing society for eventual full autonomy. Over the longer term, the convergence of electric vehicles, artificial intelligence, and shared mobility points towards cleaner, safer, and more efficient transport in a country that adds millions of new vehicles each year.

The evolution of driverless vehicles is reshaping transportation worldwide, but India's path will be distinct. Semi-autonomous systems that support human drivers are likely to take centre stage, serving as a practical bridge towards fully autonomous mobility. Equipped with AI-driven features, these systems can improve safety, reduce accidents, and optimise traffic flow.

AI-powered transportation has the potential to transform mobility in India. With sustained innovation, collaboration across industry and policy, and a clear focus on local challenges, autonomous vehicles could redefine how the country moves, making roads safer, cities smarter, and commutes more efficient. While the road ahead is undoubtedly challenging, it also offers substantial opportunities, pointing to a future in which technology and human intelligence come together to reshape transportation in one of the world's most dynamic environments. **EFY**